Lauren N. Peterson

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My goal is to improve health outcomes for people with and without disabilities by using algorithms and mathematical techniques to improve healthcare. I have experience with data-driven modeling, kinematic and dynamical systems, AI and ML algorithms, neuroengineering, and human-centered design.

Education

Rice University, Master's of Electrical & Computer Engineering — Aug 2023 – Dec 2025

- **Coursework:** AI & Systems, ML, robotics path planning and system modeling, human-centered design, algorithms, data science

University of Washington, Seattle, BS in Electrical & Computer Engineering — Sept 2018 – May 2022

- **Coursework:** Control theory, optimization techniques, neuroengineering, filter design and analysis, system design and analysis

Research Experience

Graduate Researcher, Yamagami Lab – Rice University, Houston TX — Aug 2023 – Dec 2025

- Advisor: Momona Yamagami
- Utilized computer vision and motion tracking to model upper-body gestures of people with movement disabilities
- Implemented three generative statistical models to learn the movements of participants with motor disabilities and generate novel gestures that replicate the disabled movements
- Implemented data-driven EMG algorithms and analyzed how varying the preprocessing parameters of EMG data influenced signal quality for EMG data from participants with and without motor disabilities
- Mentored two undergraduate students in best researching practices and supported one student in pursuing graduate research

Post-baccalaureate Researcher, Orsborn Lab – UW, Seattle WA — July 2022 – July 2023

- Advisor: Amy L. Orsborn
- Part of the inaugural class of postbaccalaureate researchers at the Accelerated AI Algorithms for Data-Driven Discovery (A3D3) Institute
- Used algorithms and machine learning techniques to improve the latency of closed-loop brain-computer interfaces for non-human primates
- Adapted online convex optimization algorithm to calculate the most important features in a brain-computer decoder

Undergraduate Researcher, Burden Lab – UW, Seattle WA — Dec 2018 – June 2022

- Advisor: Samuel A. Burden
- Spearheaded data collection on handedness and motor learning for 18 participants without disabilities – assisted with protocol development, recruited participants, provided informed consent, and collected tracking data on a Python GUI
- Analyzed human-subject data with control theory techniques
- Completed secondary analysis of previously-collected data – developed an algorithm to estimate the user's feedback parameters (gain and delay)

Visiting Undergraduate Researcher, LIMBS Lab – JHU, Baltimore MA — June 2021 – Aug 2021

- Advisor: Noah Cowan
- Part of the REU on Computational Science and Medical Robotics Program, Johns Hopkins University
- Collected and analyzed data from E. virescens to understand how fish learn novel tasks

- Co-authored an abstract presented at the 2022 Society of Integrative and Comparative Biology Conference [O1]

Publications

Generating Novel Representative Hand Gestures In progress.

LN Peterson, Alexandra Portnova-Fahreeva, Momona Yamagami

EMG Designing for Diversity In progress.

Claire L Mitchell, ***LN Peterson***, Chloe Park, Momona Yamagami

Online Smooth Feature Selection via Convex Smooth Optimization In progress.

Si Jia Li, ***LN Peterson***, Pavithra Rajeswaran, Leo Scholl, Amy L Orsborn

Assessing Human Feedback Parameters for Disturbance-Rejection *4th Conference on Cyber-Physical Human Systems* Dec 2022

LN Peterson, Amber H.Y. Chou, Samuel A Burden, Momona Yamagami
10.1016/j.ifacol.2023.01.094

Effect of Handedness on Learned Controllers and Sensorimotor Noise During Trajectory-Tracking *IEEE Trans. on Cybernetics* 2021

Momona Yamagami, ***LN Peterson***, Darrin Howell, Eatai Roth, Samuel A Burden

10.1109/TCYB.2021.3110187

Honors & Awards

Dean's Prize College of Engineering, Graduate Student Recruitment; Received one of two scholarships April 2023

Best Poster Award, 2nd Place; People's Choice Award A3D3 High-Throughput AI Methods and Infrastructure Workshop June 2023

Washington Research Foundation Innovation Undergraduate Fellow in Neuroengineering Nov 2020

Louis Armstrong Jazz Award June 2018

Mentorship

Steven Chen June 2024 – Aug 2024

- Summer international visiting undergraduate researcher
- Currently a Master's student at Rice University

Chloe Park Aug 2024 – May 2025

- Undergraduate researcher
- Currently a senior undergraduate student at Rice University

Oral Presentations

Towards Real-time Brain Computer Interfaces Development - Data, Algorithms, and Hardware A3D3 High-Throughput AI Methods and Infrastructure Workshop June 2023

LN Peterson, Si Jia Li, Amy L Orsborn

Can a Fish Learn to Ride a Bicycle? Johns Hopkins CSMR-REU Final Presentations Aug 2021

LN Peterson, Yu Yang, Dominic Yared, Noah Cowan

Effects of Dominant and Nondominant Hands on Feedforward and Feedback Adaptations, UW Undergraduate Research Symposium April 2020

LN Peterson, Momona Yamagami, Samuel A Burden

Poster Presentations

Generating Novel Representative Gestures from Kinematics Data. 2024 TEROS, College Station, TX. May 2024

LN Peterson, Momona Yamagami

Using convex feature selection to improve offline movement decoding A3D3 High-Throughput AI Methods and Infrastructure Workshop June 2023

LN Peterson, Si Jia Li, Leo R Scholl, Pavithra Rajaswaran, Lydia I Smith, Amy L Orsborn

Using convex feature selection to improve offline movement decoding Neuroengineering Research Symposium May 2023

LN Peterson, Si Jia Li, Leo R Scholl, Pavithra Rajaswaran, Lydia I Smith, Amy L Orsborn

Effects of Dominant and Nondominant Hands on Feedforward and Feedback Adaptations UWIN Fellowship Research Symposium Jan 2020

LN Peterson, Momona Yamagami, Samuel A Burden

Technologies

Languages: Python, Matlab, Java, C++, R, SQL

Prototyping: 3D printing, woodworking, sautering

References

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